NBA opening-spread walk-forward — model and strategy review

Strategy: Directional value-taking against the opening point spread in NBA regular-season games. A market-blind margin model prices every listed game; its disagreement with the opening spread is then spent, at about a third of face value, as a correction *to* the opening spread — `m_final = open_margin + f(x)`, with `f` a ridge shrunk hard toward zero and fitted walk-forward. A single side is taken where the corrected margin still disagrees with the line by more than a walk-forward selected threshold, at the best number available across books. Every position is one side of one game at –110, entered at the open and held to settlement: no hedge, no offsetting leg, no in-game management, no exit. Flat 1 unit per bet, so nothing compounds. No calendar filter.

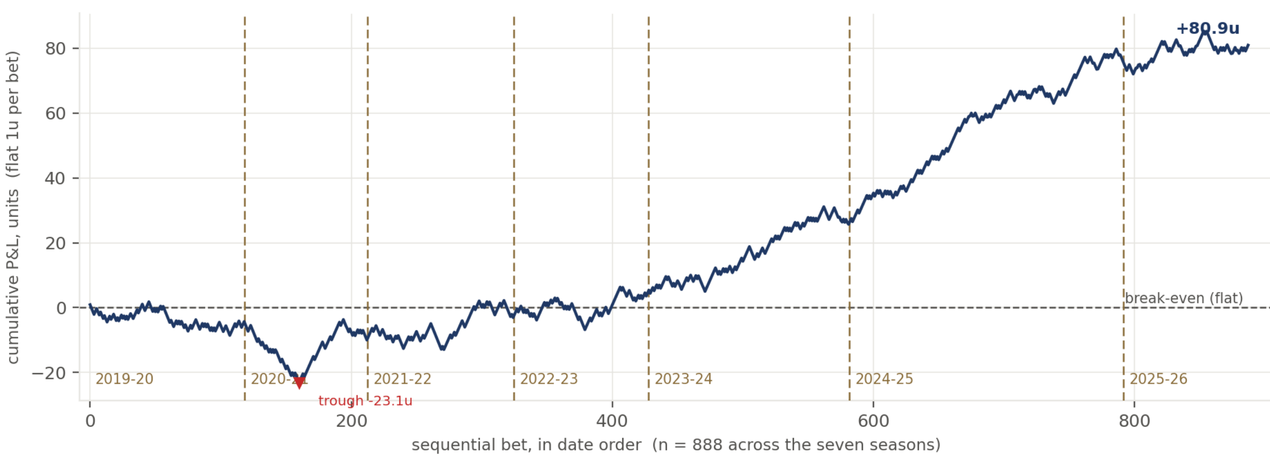
Selection is the strategy. The book is roughly 10% of the slate; the cover rate on the selected book is not the cover rate on the slate.

Frame: 888 bets, 7 seasons, 2019-20 → 2025-26. Earlier seasons are excluded on principle: the daily injury report begins 2018-12-17 and the availability leg is half the production margin, so before 2019-20 the model cannot run as designed. The configuration is selected on seasons 1..k, frozen, scored on k+1, and rolled forward; the selection history reaches back to 2012-13.

Headline results

season	books/game at open	bets	P&L (u)	ROI	Sharpe (ann.)
2019-20	—	119	–4.28u	–3.60%	–0.45
2020-21	—	94	–5.71u	–6.07%	–0.60
2021-22	—	112	+6.93u	+6.18%	0.63
2022-23	—	103	+7.55u	+7.33%	0.90
2023-24	7.74	154	+21.20u	+13.77%	1.90
2024-25	1.00	210	+50.40u	+24.00%	3.81
2025-26	1.03	96	+4.85u	+5.05%	0.54
2019-26 pooled, best of 9 books	—	**888**	**+80.93u**	**+9.11%**	**1.10**
2019-26 pooled, after outlier haircut	—	888	+69.56u	+7.83%	—

ROI is P&L per unit staked. Sharpe is annualized by √(sessions per season), not √252, which would inflate it 1.8×. Prices are the best of 9 books, but that panel is measured only in 2023-24, so the multi-book figures are modelled for every other season. The haircut row charges for the 8.1% of best-of-N prices that sit more than 1.5 points off the next book.



All 888 bets in date order at the opening spread and –110; flat 1u, so the path is a running sum and nothing is re-selected within it. Trough is –23.1u at bet 161. The first three seasons lose money; the last four make +83.5u.

Pooled ROI is **+9.11%** over 888 bets, with a season-clustered 95% interval of **[–2.75%, +16.08%]** and an MDE80 of 12.7pp. **The interval contains zero** and 2024-25 alone supplies **62%** of the P&L. The frame is also too short to tune on: measured, a null taking the best of five randomly chosen game subsets buys **+2.54 ROI points** on average, and every strategy filter tested lands inside that band. Shortening the window does not help — the best 4-season window returns +14.92% against a +9.33% average across all 4-season windows, so **+5.6 of those points are the selection, not a regime.**

The model

The model outputs a margin, not a probability. A spread is itself a margin forecast, so the sides test involves no devig or probability conversion.

```
margin = 0.5·four_factors + 0.5·availability_composition + schedule_layer + tank_term
P(home win) = sigmoid(margin / 7.2)
```

Two independent estimates of team strength, averaged, plus additive context. The market-blind model never sees market odds, enforced structurally rather than by convention; only the offset layer above it sees the price.

component	what it is	how it is fit
Four factors	opponent-adjusted ratings on shooting, turnovers, rebounding and free-throw rate; a decomposition of points per possession rather than a feature shortlist	one L2 ridge solve per factor (`ridge=25`), then the four adjusted values mapped to points by a fitted linear map. An 8-factor extension tied, so 4 is kept for parsimony
Availability composition	Σ over available players of DARKO talent × trailing minutes / 48; the leg that reacts to injuries	each player weighted by 1 – P(out), forecast from as-of-open information only. Requires the daily injury report; this is why the frame starts 2019-20
Schedule layer	home edge, back-to-backs, dead-team flags	estimated walk-forward, EB shrinkage `n/(n+600)` toward a prior, `team_home_ridge=200` on per-team home deviations. The only component that has survived strict out-of-sample testing on every split tried
Tank term	late-season effort	exactly zero outside its window
Offset layer	the correction to the opening line	ridge on three features knowable at the open, shrunk hard toward zero. Fitted edge coefficient 0.33–0.37 in every fold

How it compares to the market

Model, opening line and closing line scored on identical games:

source	log loss	
market-blind model	0.59276	beaten by both
opening line	0.59228	
offset construction	0.58865	recovers 26.7% of open→close
closing line	0.57870	

The market-blind model does not beat the opening line standing alone — it is 0.00048 worse than the open and 0.01406 worse than the close. It is not a discarded attempt: it is the offset layer's dominant input, and the finding that its disagreement must be spent at a third of face value against a market anchor, rather than trusted on its own, is the strategy.

Notes and caveats

- Results are **simulated** on recorded opening prices with a walk-forward selection rule. **No capital has ever been deployed.**
- Multi-book pricing is measured in 2023-24 only and modelled elsewhere.
- The architecture was developed on 2021-26 data, so retrospective performance is not an untouched test. **2026-27 is the first decisive prospective evaluation.**
- Neither arm is the production default; promoting one is a re-certification.
- Full method, the register of 209 decisions (most of them rejections), the bet-time information leak and its fix, and the window-selection analysis are in the repository README and `docs/`.